

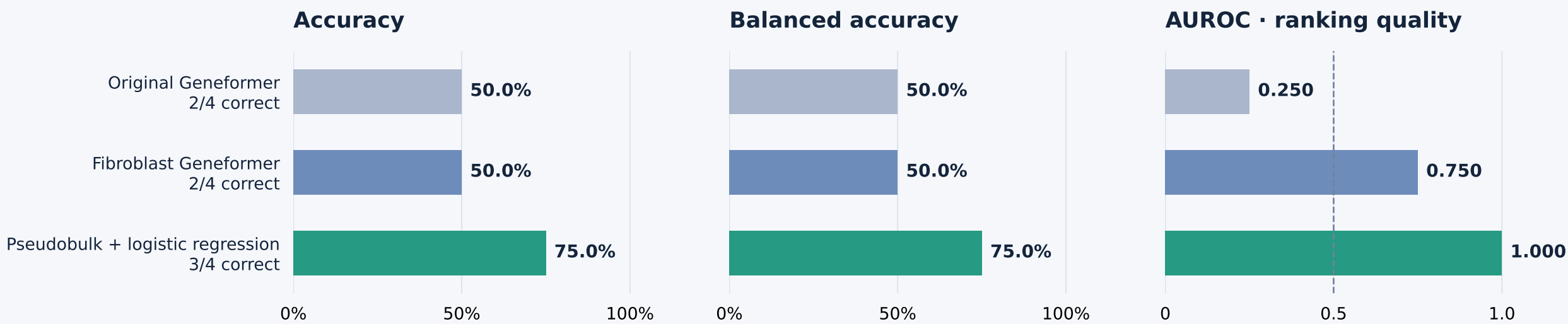
genepilot

Recognise senescent cells from gene activity and propose genes for laboratory research.



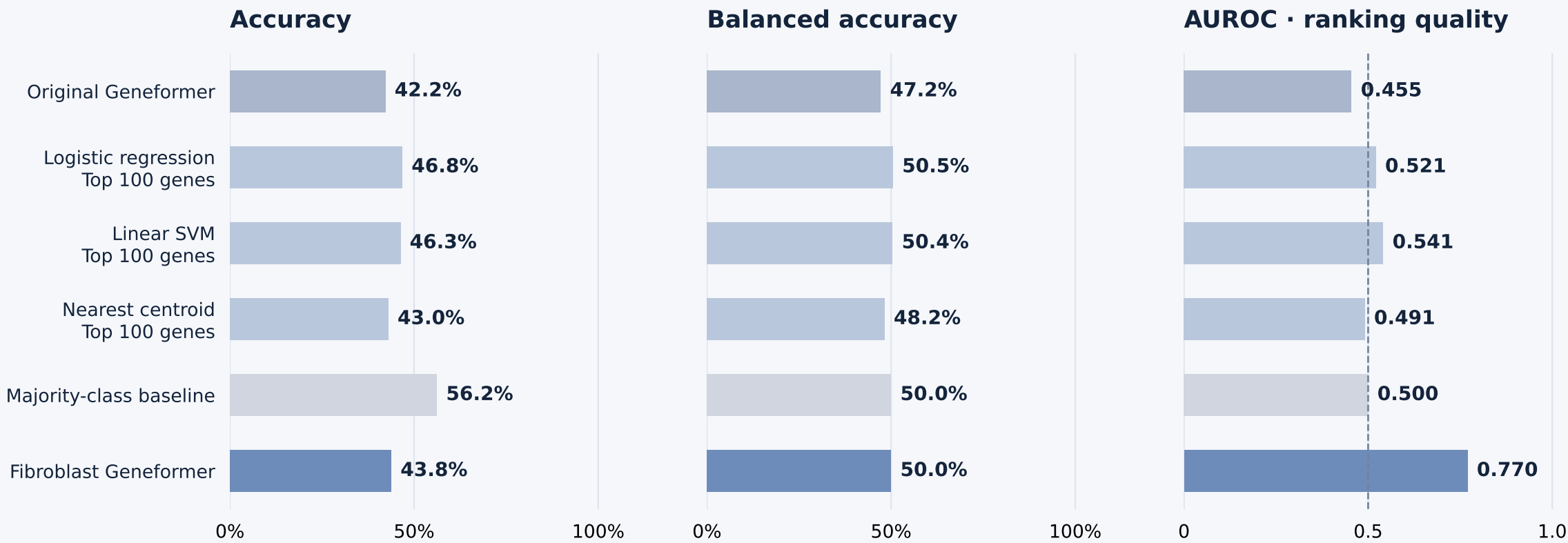
Sample-level results · 4 external samples

Same GSE282425 benchmark. Geneformer averages cell predictions; pseudobulk combines raw counts first.



Cell-level results · 3,097 external cells

Original models trained on 70 mixed lung cells; fibroblast Geneformer trained on 14,000 fibroblasts.



Earlier classical study (GSE63577): 16/20 reference markers recovered; reported accuracy 76.7%.

That accuracy used excluded cell lines within one study; it is not directly comparable to the external results above.

Accuracy = fraction correct. Balanced accuracy = average success across the two classes.

AUROC = ranking quality (0.5: chance; 1.0: perfect ordering). AUROC 1.0 does not mean 100% accuracy.

Preliminary: the 4 external samples come from one donor. Pseudobulk has no individual-cell predictions.

Candidate genes require laboratory validation. More independent samples are needed to confirm performance.

Sources: GSE282425 REPORT.md, FIBROBLAST_OPTION1_REPORT.md, PSEUDOBULK_REPORT.md; EVALUATION.md; report.md.